X-ray project Report



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Table of contents

Inhoud

Revision history 3
1. Introduction 4
2. Sprint 1 5
   2.1 Context diagram 6
   2.2 Use case diagram 7
   2.3 Module structure diagram 9
3. Sprint 2 12
   3.1 PC state diagram 13
   3.2 CentralAcquisition state diagram 14
   3.3 CentralAcquisition context diagram 15
   3.4 CentralAcquisition use case diagram 16
   3.5 CentralAcquisition module structure diagram 18
4. Sprint 3 19
   4.1 Geometry context diagram 20
   4.2 Geometry use case diagram 21
   4.3 Geometry state diagram 23
   4.4 X-ray generator context diagram 24
   4.5 XrayGenerator use case diagram 25
   4.6 XrayGenerator state diagram 27

Revision history

Revision date	Changes made
22-09-2025	Started on the document, put in the introduction, sprint summary and context diagram
26-09-2025	Finished the first sprint of the document with the use case diagram and module structure diagram
21-10-2025	Reworked the diagrams based on the feedback that was given
29-10-2025	Added the state diagrams and the module structure diagram of the CentralAcquisition
30-10-2025	Reworked the state diagrams with the feedback I got from them
31-10-2025	Made the context diagrams and use-case diagram for the CentralAcquisition, got feedback and implemented the feedback for that
13-11-2025	Fixed some small mistakes within the state diagram of the CentralAcquisition when reviewing it together with Martyn after he got his feedback
20-10-2025	Made the context diagrams, use case diagrams and use cases and the state diagrams for the geometry and x-ray generator modules. Also got feedback and used the feedback to better these diagrams

1. Introduction

In this report the process of the x-ray project will be documented. This progress will be segmented in sprints. When something changes of something made in an earlier sprint, the changed version will be put in the current sprint and not be changed in the previous sprint to show the things that were changed.

2. Sprint 1

In this sprint there were multiple things done. Most of these things were start to code the project, but there were also some exercises that had to be done to visualize the project some more. These exercises were to make:

- A context diagram
- A use-case diagram
- A module structure diagram

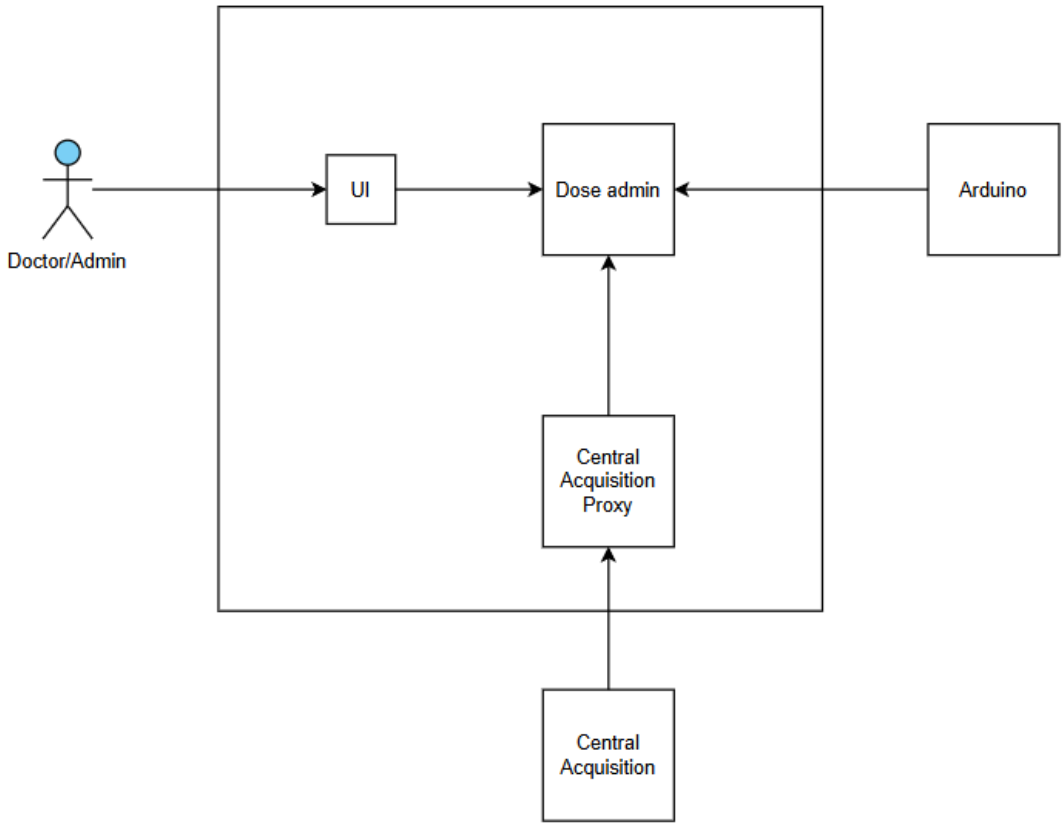
These three diagrams are visible inside this part of the report.

2.1 Context diagram

A context diagram is a diagram where the context of the system is being visualized and how the different sub-systems will interact with each other.

For the x-ray system the following components are used in the complete system:

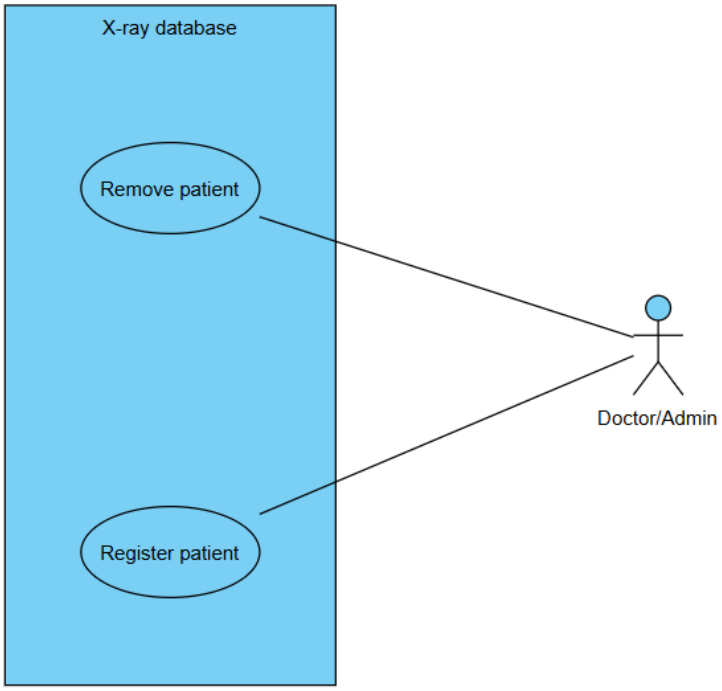
- Doctor/Admin: This is the user of the database
- UI: This is the user interface creates the visuals of the database so the database can be used
- Dose admin: This is the shared code that every part of the console communicates to
- Arduino: This is the hardware of the system that communicates with the console
- Central Acquisition: This defines what gets send to the system and what it will mean
- Central Acquisition Proxy: Makes sure that the whole system gets started up correctly.



Picture 1: the Context diagram of the whole system

2.2 Use case diagram

A use case diagram is a way of visualizing the interactions between the users and the system and what the user is allowed to do within the system.



Picture 2: A use-case diagram for the console application in the x-ray machine

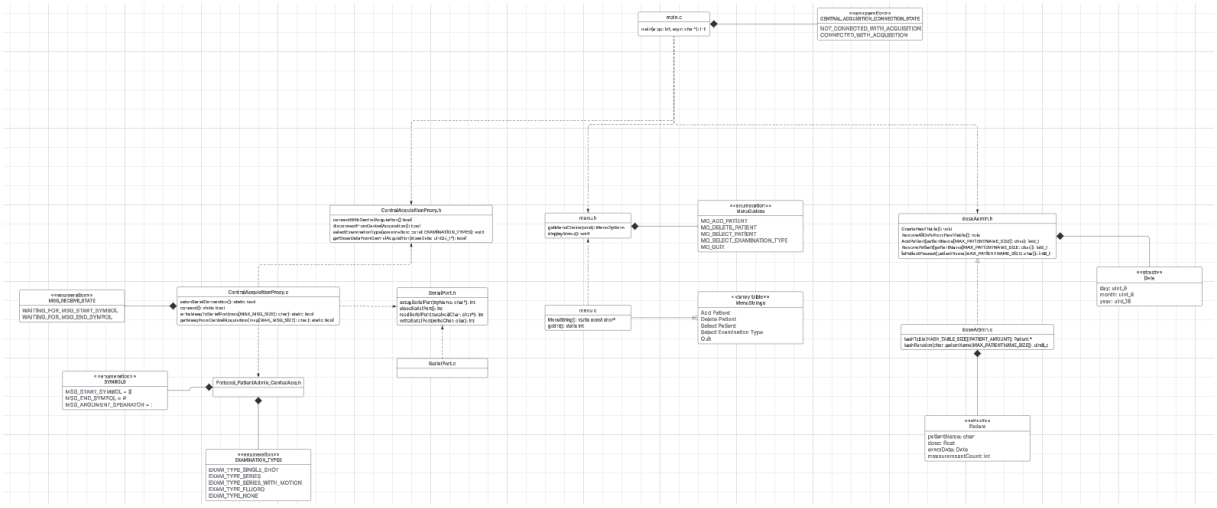
For this use-case diagram there are a couple use-case descriptions which are described on the next page

Use-case ID: UC_001	**Use-case:** registering patient
Description	The actor adds a new patient into the database
Actor:	Doctor/Admin
Flow:	1. Actor enters name of patient 2. System checks the name 3. System gives feedback that name already exists 4. Else system gives feedback that patient exceeds MAX_PATIENTNAME_SIZE 5. Else system adds name to database

Use-case ID: UC_002	**Use-case:** Removing patient
Description	The actor removes an existing patient from the database
Actor:	Doctor/Admin
Flow:	1. Actor enter name of patient 2. System checks the name 3. System gives feedback that name doesn't exist already 4. Else system gives feedback that patient exceeds MAX_PATIENTNAME_SIZE 5. Else system sets all of the data connected to that person to NULL

2.3 Module structure diagram

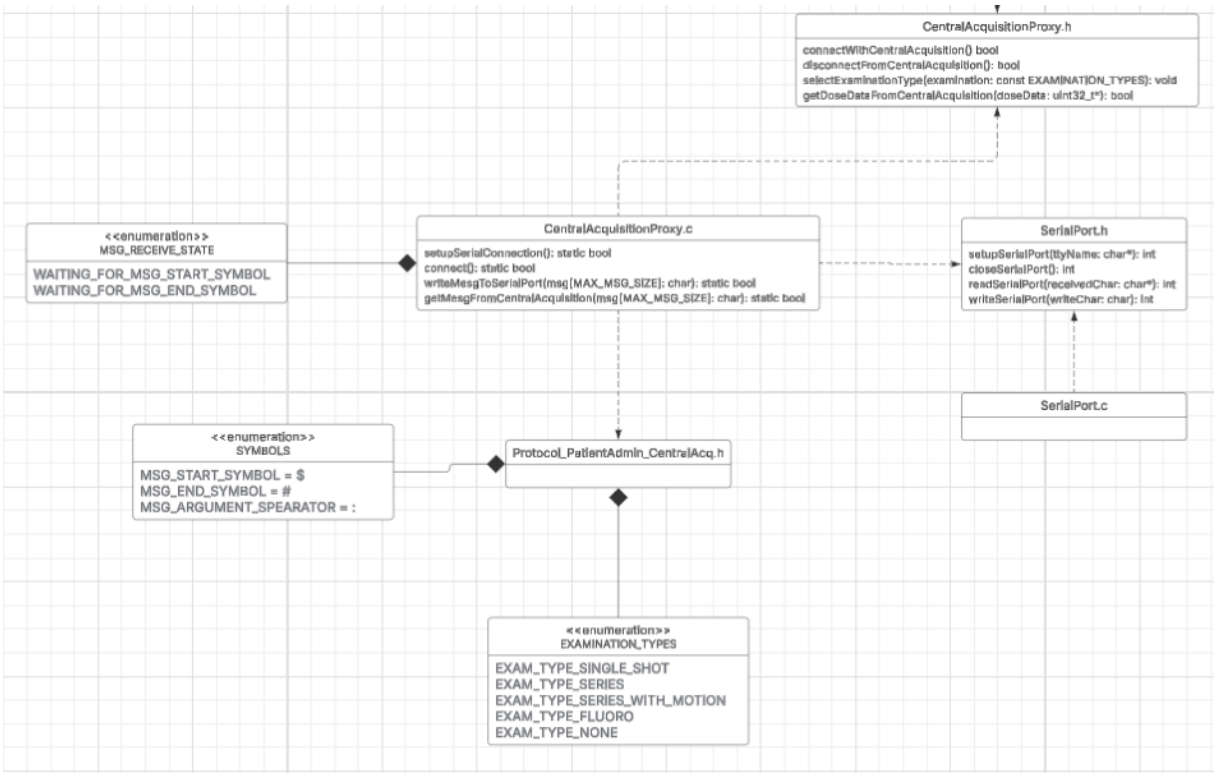
A module structure diagram is a way of visualizing a complex system. This way there will always be a way to see what part of the system does what function and where specific data may come from.
Because the total diagram is to big to show it in on one page it will be shown, but after that the diagram will be separated into four sections.



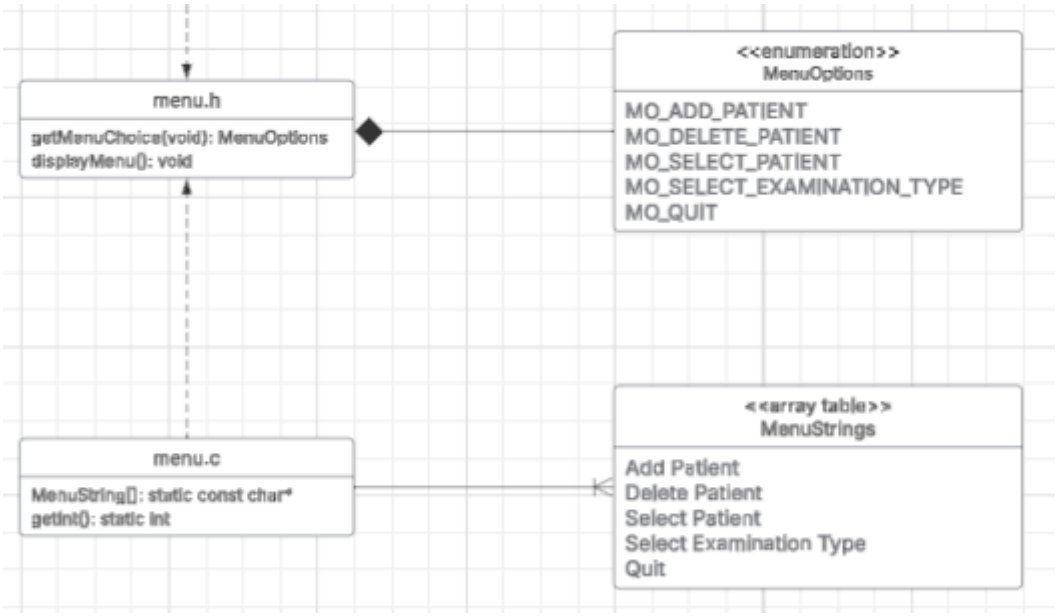
Picture 3: A module structure diagram for the console application in the x-ray machine



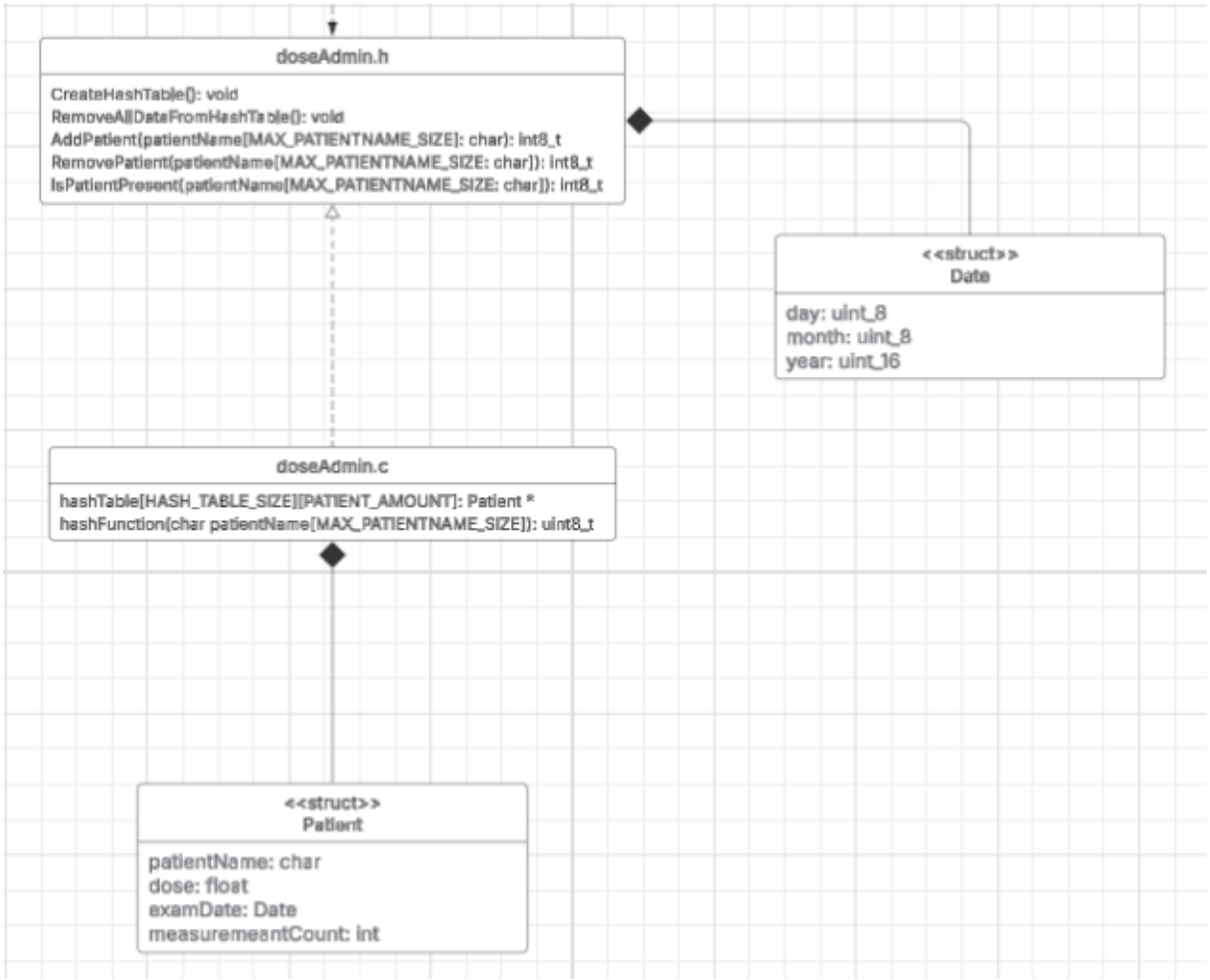
Picture 4: In this picture the section of the structure diagram that includes main.c is visible.



Picture 5: In this picture the section of the structure diagram that includes CentralAcquisitionProxy.h is visible. CentralAcquisitionProxy.h gets included by main.c



Picture 6: In this picture the section of the structure diagram that includes menu.h is visible. menu.h gets included by main.c



Picture 7: In this picture the section of the structure diagram that includes doseAdmin.h is visible. doseAdmin.h gets included by main.c

3. Sprint 2

In this sprint multiple diagrams had to be constructed. To start two different state diagrams had to be made for the PC and the CentralAcquistition. Besides that the CentralAcquistition also got the three diagrams made for the DoseAdmin in the last sprint.

- Context diagram
- Use case diagram
- Module structure diagram

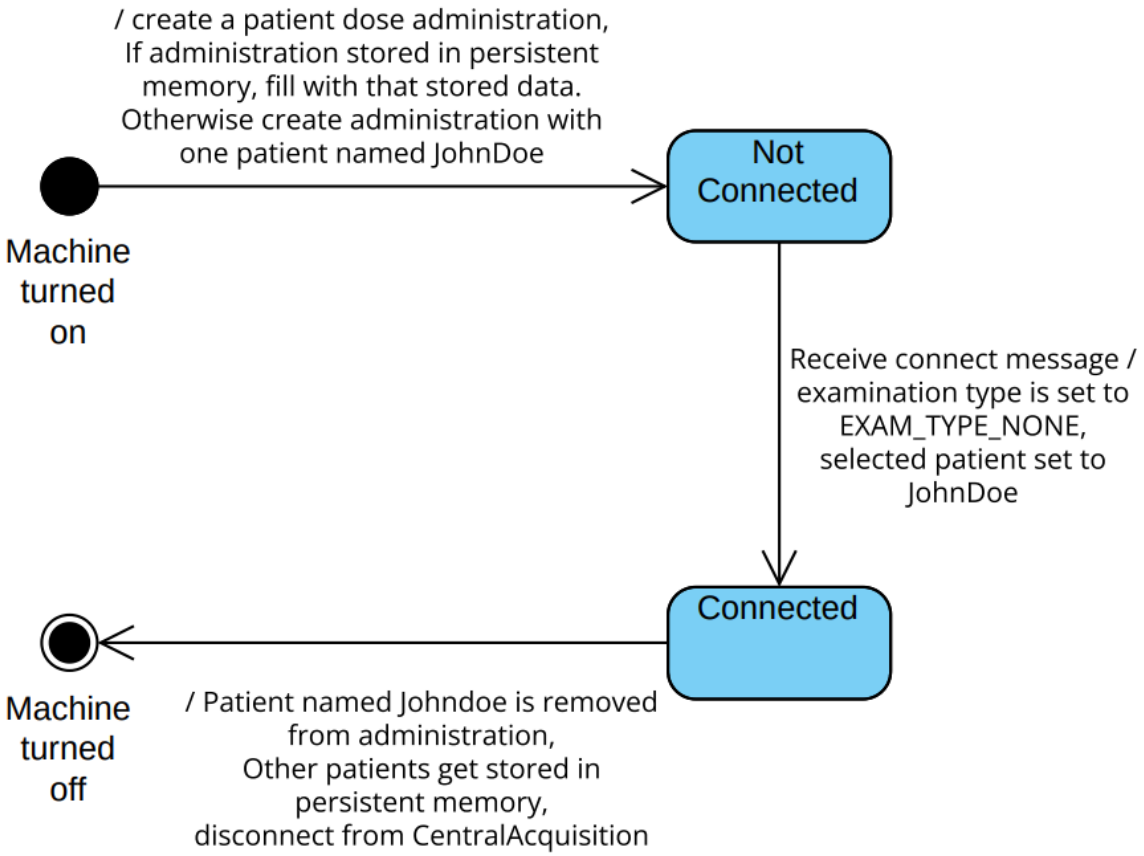
All of these diagrams will be visible in this part of the report.

3.1 PC state diagram

The paths in which the states get there are described. First with the descriptions and after that the actions, these two are divided with a / so if the text starts with a / it immediately starts with the actions. The beginning and ending of a state diagram are visible with a full black dot for the beginning and a black dot with a black circle encircling it is used to show the ending of the state diagram

In the state diagram of the PC the different states it can go in are visible. These state are:

- not connected
- connected



Picture 8: In this picture the state diagram of the PC is visible

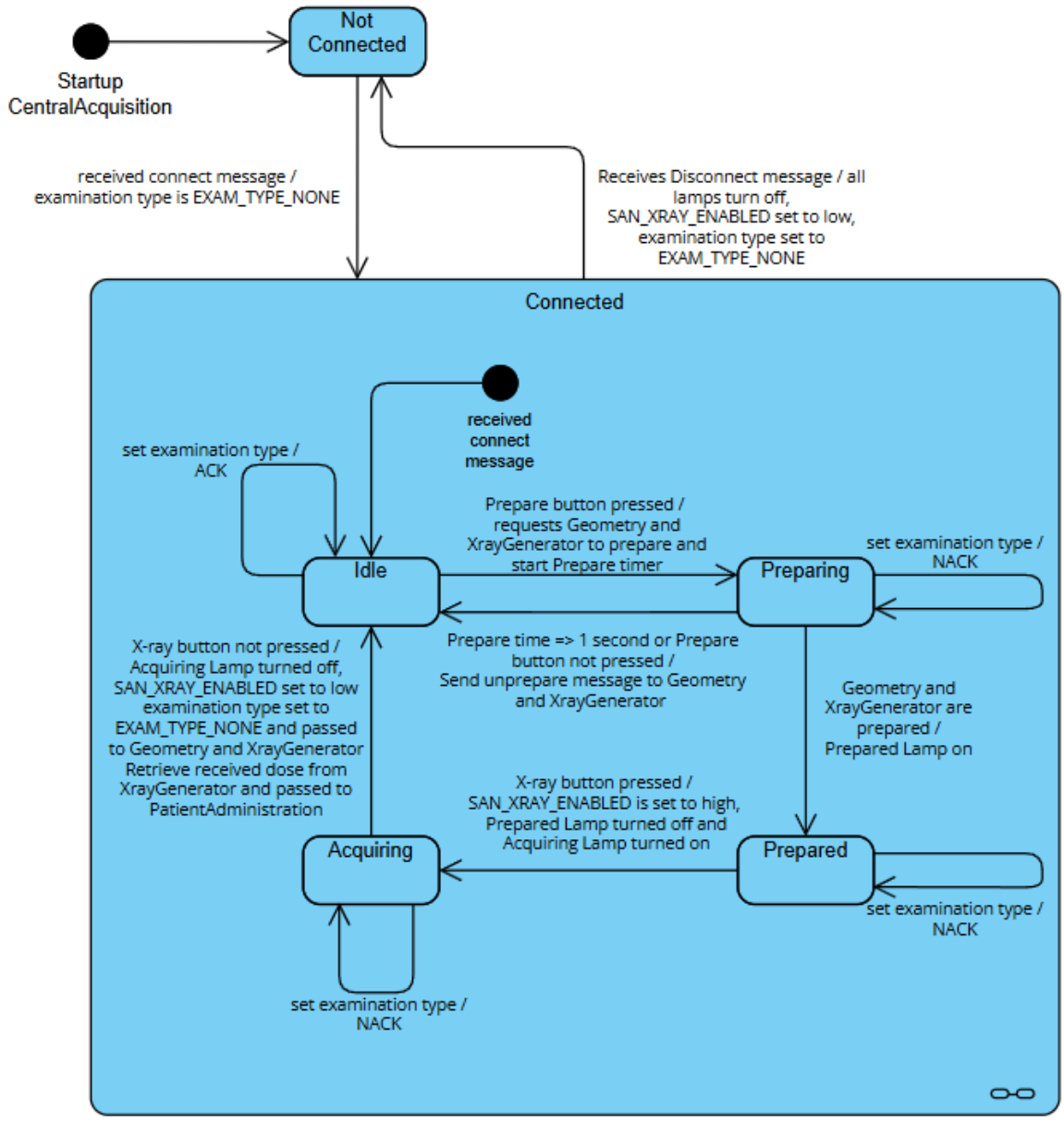
3.2 CentralAcquisition state diagram

In the state diagram for the CentralAcquisition the different states it can go in are visible. These state are:

- not connected
- connected

Connected has a couple of substates inside of it, which are:

- Idle
- Preparing
- Prepared
- Acquiring

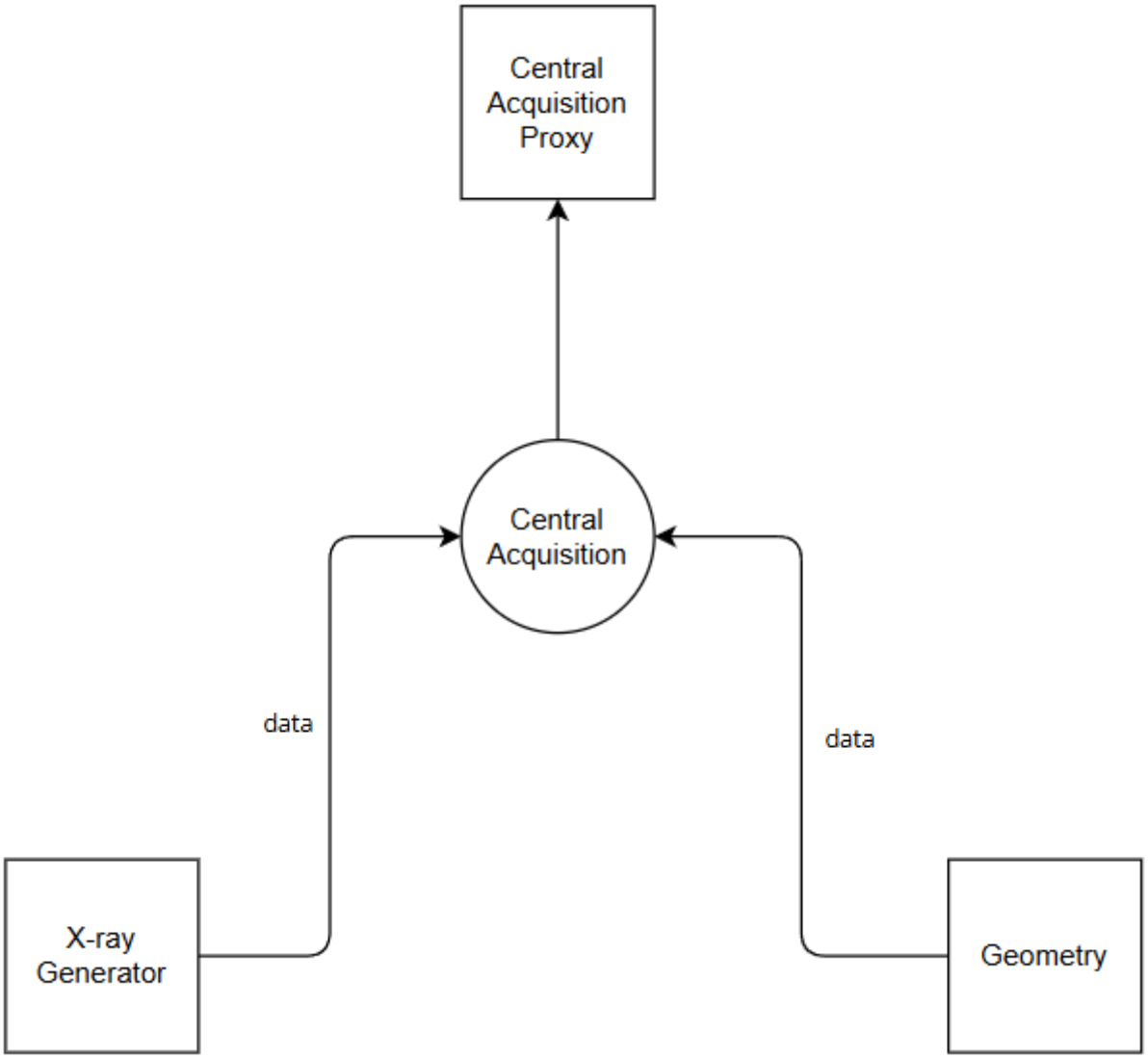


Picture 9: In this picture the state diagram of the CentralAcquisition is visible

3.3 CentralAcquisition context diagram

For the CentralAcquisition the following components are used:

- Geometry: This gives the relative position of the person to the X-ray generator
- X-ray Generator: This generates the X-ray that will be absorbed by the patient
- CentralAcquisitionProxy: This is the bridge between the hardware and the PC so that the commands actually work

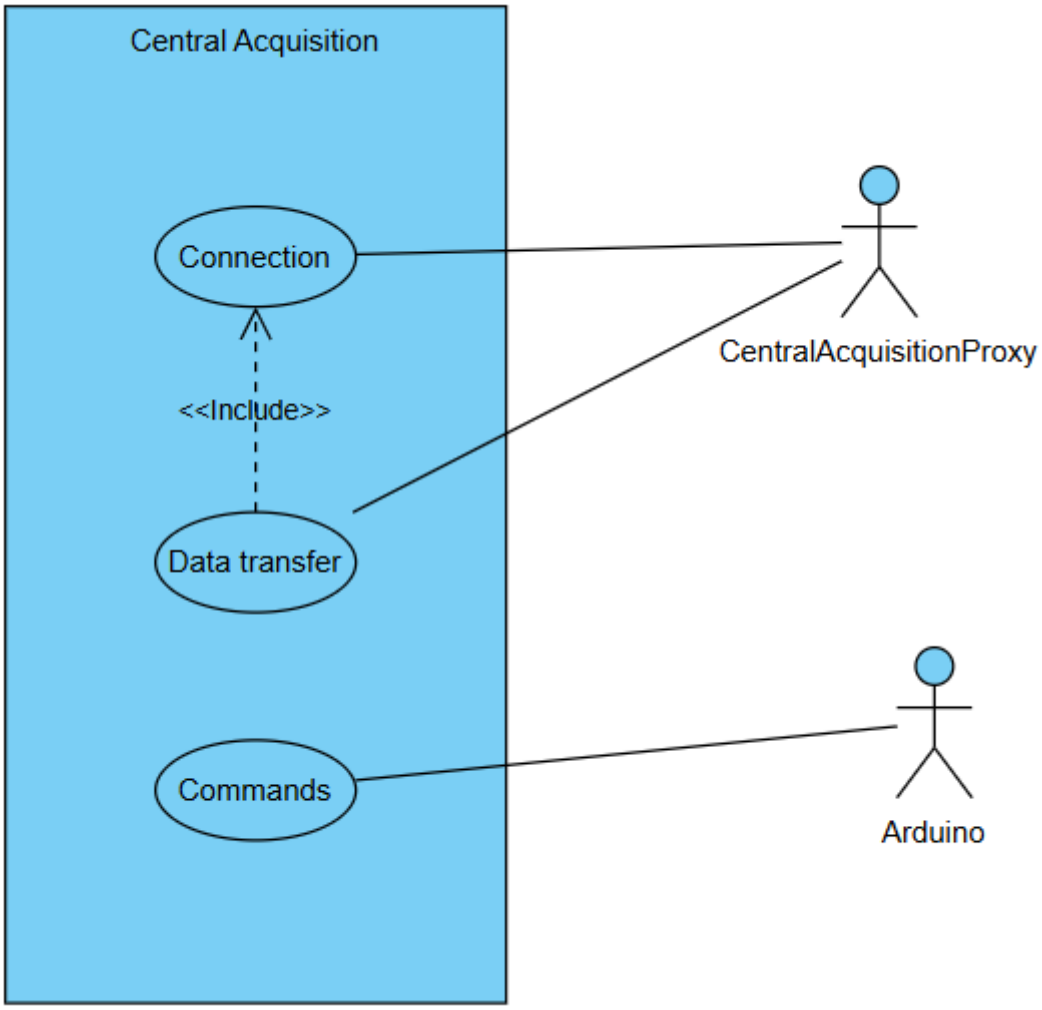


Picture 10: In the picture above the context diagram of the CentralAcquisition is shown

With this system there are a couple of subsystems that have to be made, these context diagrams will be made and visible in sprint 3.

3.4 CentralAcquisition use case diagram

In the picture that will be visible below this the use case diagram of the CentralAcquisition will be visible. After that the different use cases will be described



Picture 11: In the picture above the use case diagram of the CentralAcquisition is visible

For this use-case diagram there are a couple use-case descriptions which are described on the next page

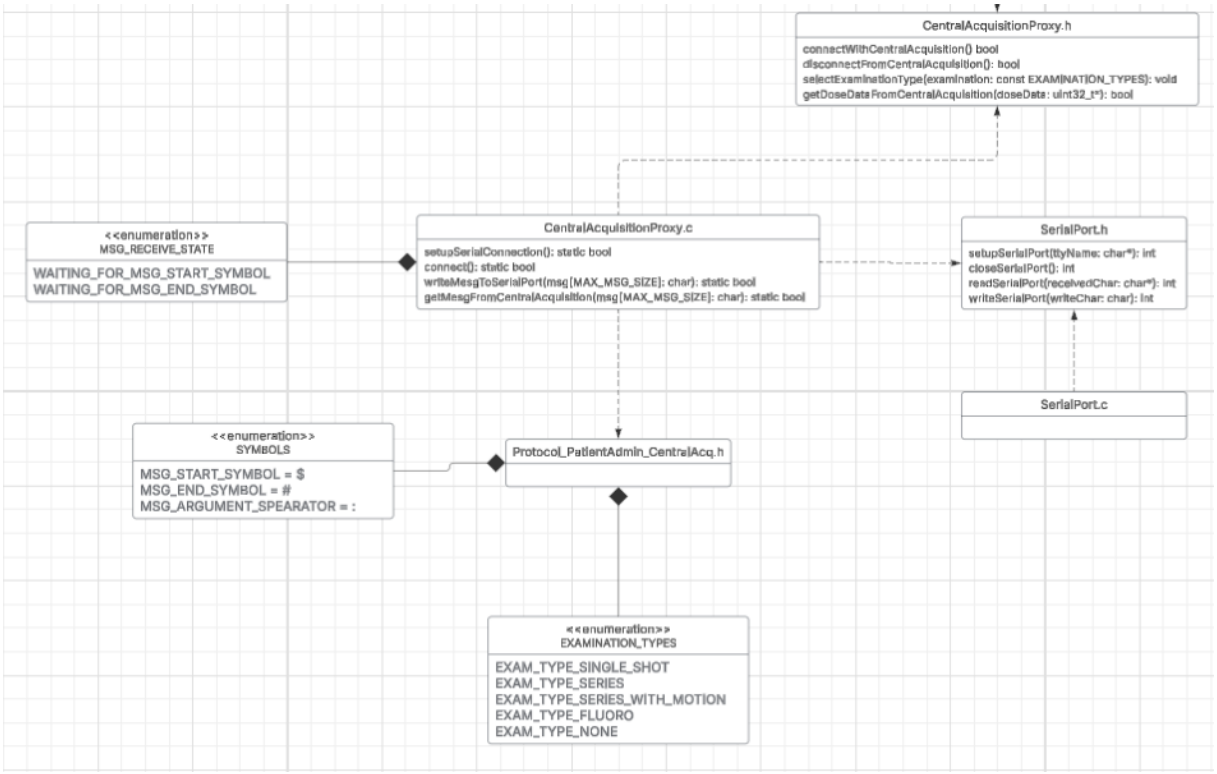
Use-case ID: UC_101	**Use-case:** Connection
Description	The actor makes connection with the CentralAcquisition
Actor:	CentralAcquisitionProxy
Flow:	1. The actor tries to connect to the CentralAcquisition 2. The actor gets a fail message that connection wasn't possible 3. Else the CentralAcquisition connects to the actor

Use-case ID: UC_102	**Use-case:** Data transfer
Description	The actor transfers or obtains data from the CentralAcquisition
Actor:	CentralAcquisitionProxy
Flow:	1. Actor asks to transfer data to or from the CentralAcquisition 2. The system says it is not connected to the actor 3. Else the data transfer doesn't go through because there is no data or no valid data type to transfer 4. Else the CentralAcquisition transfers data to or from the actor

Use-case ID: UC_103	**Use-case:** Commands
Description	The actor executes a command coming from the CentralAcquisition
Actor:	Arduino
Flow:	1. The CentralAcquisition requests the actor to execute a command 2. The actor executes the command 3. The actor gives an error that the command went wrong somewhere 4. Else the actor give the obtained data from the command

3.5 CentralAcquisition module structure diagram

Under this the module structure diagram of the CentralAcquisition module is visible. In this diagram the different functions get place in the part of the module where they get called and which different programs communicate or use information from each other



Picture 12: In this picture the module structure diagram of the CentralAcquisition is visible.

4. Sprint 3

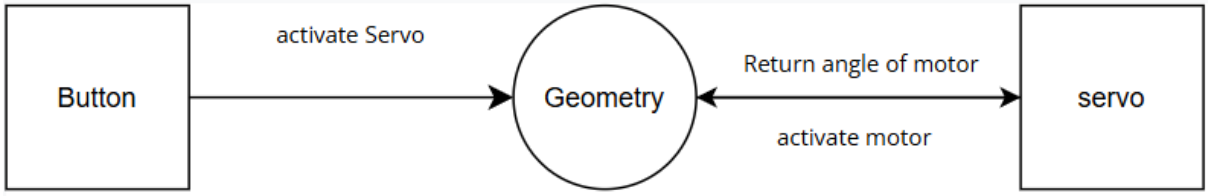
In this sprint multiple diagrams had to be constructed again. All these diagrams were connected to the functions Geometry and XrayGenerator. The diagrams made for Geometry and XrayGenerator were:

- Context diagram
- Use case diagram
- State diagram

All of these diagrams are visible in the next part of the report

4.1 Geometry context diagram

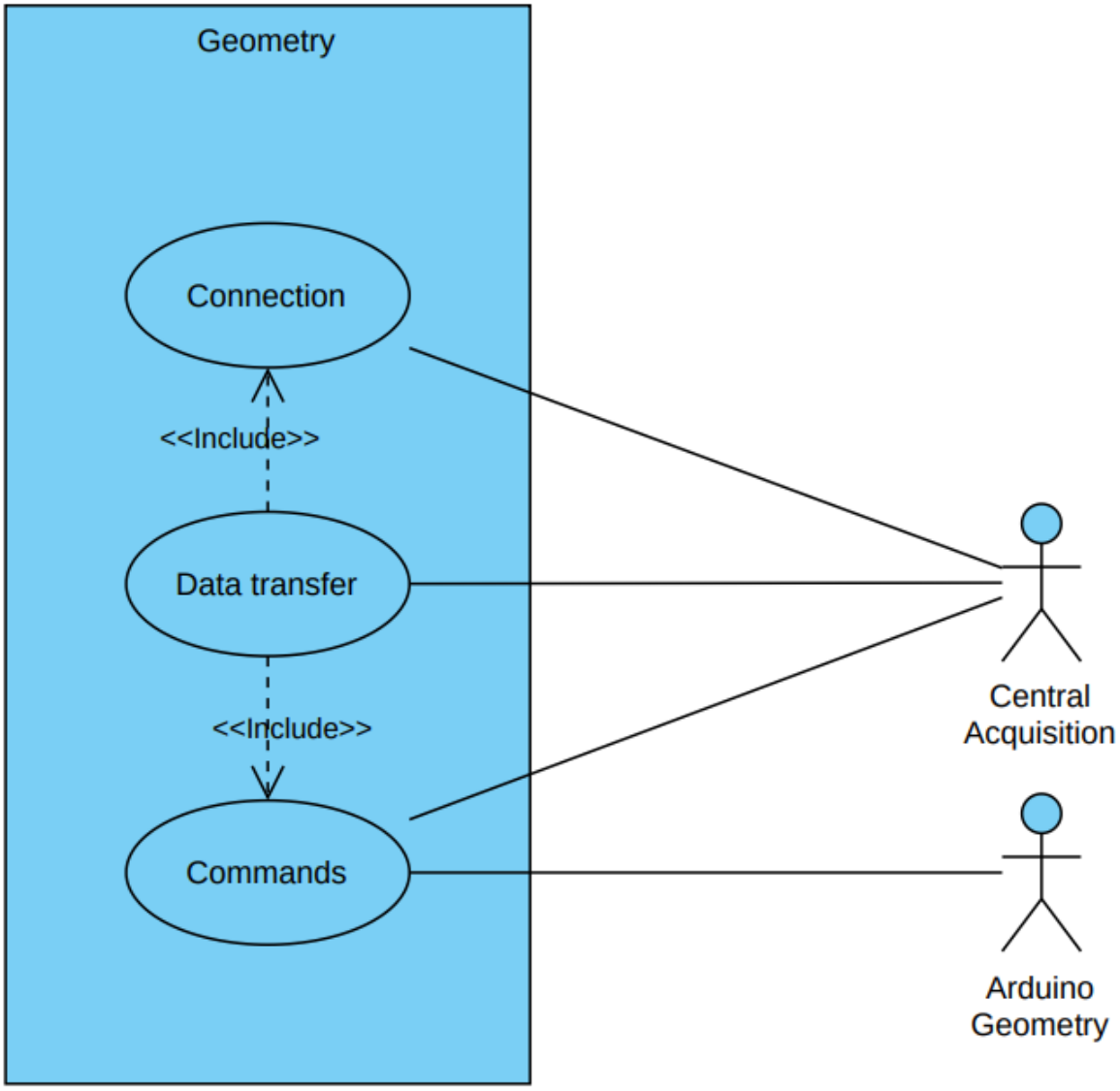
The geometry is meant to return the angle at which the X-ray generator is placed relative to the patient who is in the X-ray machine. This moving is done with a button the lets the servo move in a set speed and the servo returns the value of the angle it is stood. The context diagram of the geometry is shown below:



Picture 13: In the picture above the context diagram is shown of the geometry subsystem

4.2 Geometry use case diagram

In the picture that will be visible below this the use case diagram of the Geometry module will be visible. After that the different use cases will be described



Picture 14: In the picture above the use case diagram of the geometry module is visible

For this use-case diagram there are a couple use-case descriptions which are described on the next page

Use-case ID: UC_201	**Use-case:** Connection
Description	The actor makes connection with the geometry module
Actor:	CentralAcquisition
Flow:	1. The actor tries to connect to the geometry module 2. The actor gets a fail message connecting wasn't possible or failed 3. Else the geometry module connects to the actor

Use-case ID: UC_202	**Use-case:** Data transfer
Description	The actor obtains data from the geometry module
Actor:	CentralAcquisition
Flow:	1. Actor asks to obtain data from the geometry module 2. The geometry module says it is not connected to the actor 3. Else the geometry module says no command was executed that could have generated data 4. Else the geometry module says the data is not a valid data type to be transferred 5. Else the geometry module transfers data to or from the actor

Use-case ID: UC_203	**Use-case:** Commands
Description	The geometry module executes a command coming from the actor
Actor:	CentralAcquisition
Flow:	1. The actor requests the geometry module to execute a command 2. The geometry module gives an error that the command went wrong somewhere 3. Else the geometry module executes the command given by the actor

Use-case ID: UC_204	**Use-case:** Commands
Description	The geometry module executes a command coming from the actor
Actor:	Geometry arduino
Flow:	1. The actor requests the geometry module to execute a command 2. The geometry module gives an error that the command went wrong somewhere 3. Else the geometry module executes the command given by the actor

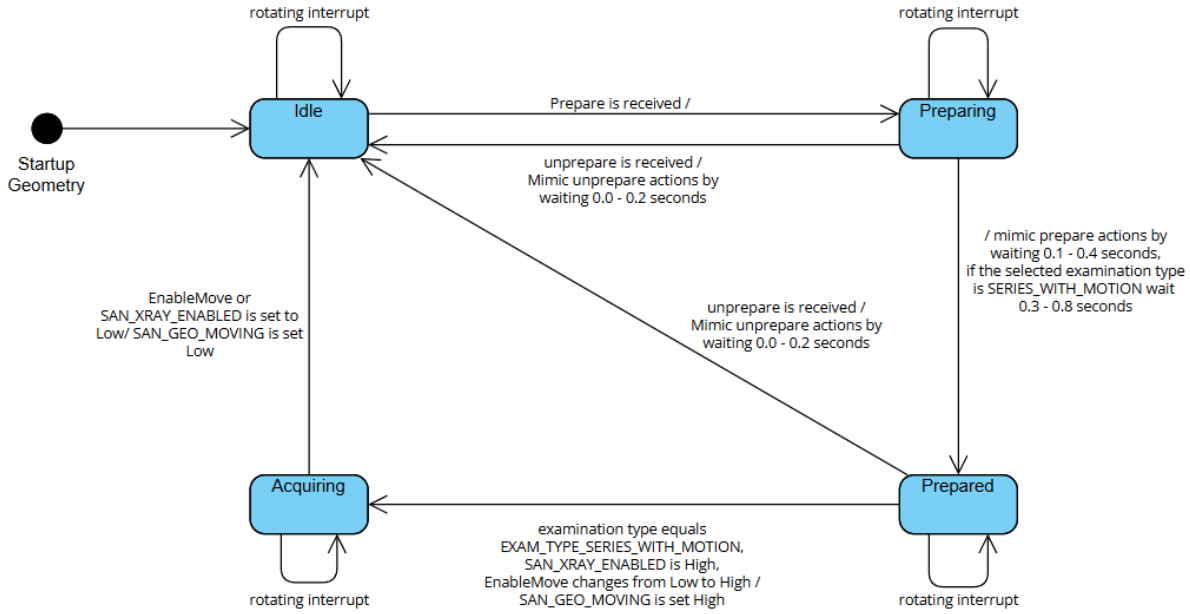
4.3 Geometry state diagram

In the state diagram for the geometry module the different states it can go in are visible. These state are:

- Idle
- Preparing
- Prepared
- Acquiring

The way these states can change into each other and what happens when changing to the different states is shown in the state diagram below

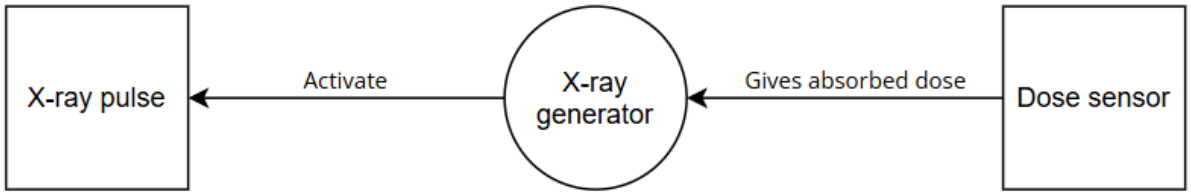
Interrupt means: / When motor is not turning and joystick or the FixedSpeed button is pressed the motor will start to rotate, when motor is turning because of the FixedSpeed button and FixedSpeed button is pressed again it will stop rotating



Picture 15: In this picture the state diagram of the Geometry module is visible

4.4 X-ray generator context diagram

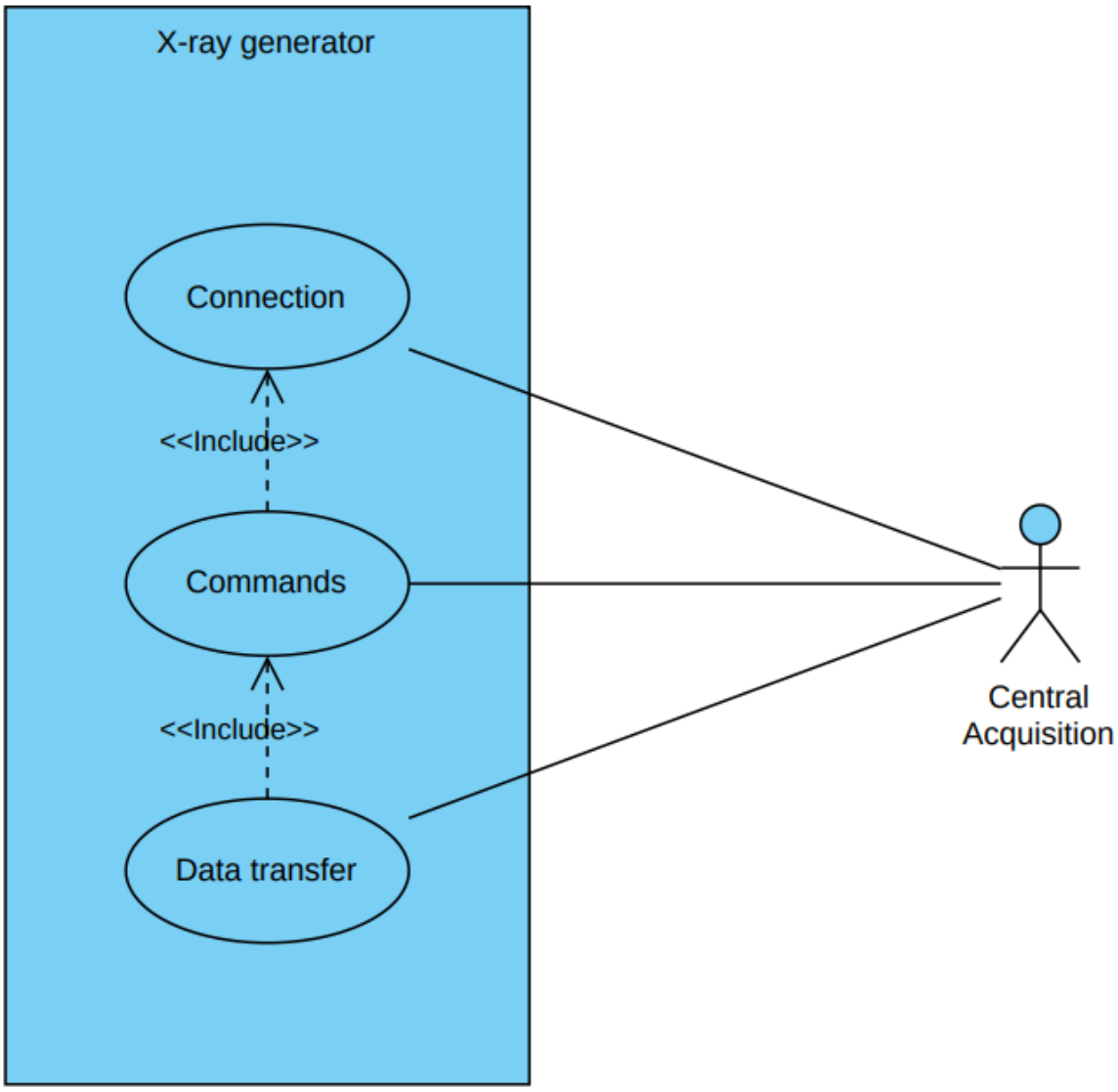
The X-ray generator is meant to send out an x-ray pulse and return the dose absorbed by the patient inside the machine. The context diagram of the X-ray generator is shown below:



Picture 16: In the picture above the context diagram is shown of the X-ray generator subsystem

4.5 XrayGenerator use case diagram

In the picture that will be visible below this the use case diagram of the X-ray generator module will be visible. After that the different use cases will be described



Picture 17: In the picture above the use case diagram of the X-ray generator module is visible

For this use-case diagram there are a couple use-case descriptions which are described on the next page

Use-case ID: UC_301	**Use-case:** Connection
Description	The actor makes connection with the X-ray generator module
Actor:	CentralAcquisition
Flow:	1. The actor tries to connect to the X-ray generator module 2. The actor gets a fail message connecting wasn't possible or failed 3. Else the X-ray generator module connects to the actor

Use-case ID: UC_302	**Use-case:** Commands
Description	The X-ray generator module executes a command coming from the actor
Actor:	CentralAcquisition
Flow:	1. The actor requests the X-ray generator module to execute a command 2. The X-ray generator module says it is not connected to the actor 3. Else the X-ray generator module gives an error that the command went wrong somewhere 4. Else the X-ray generator module executes the command given by the actor

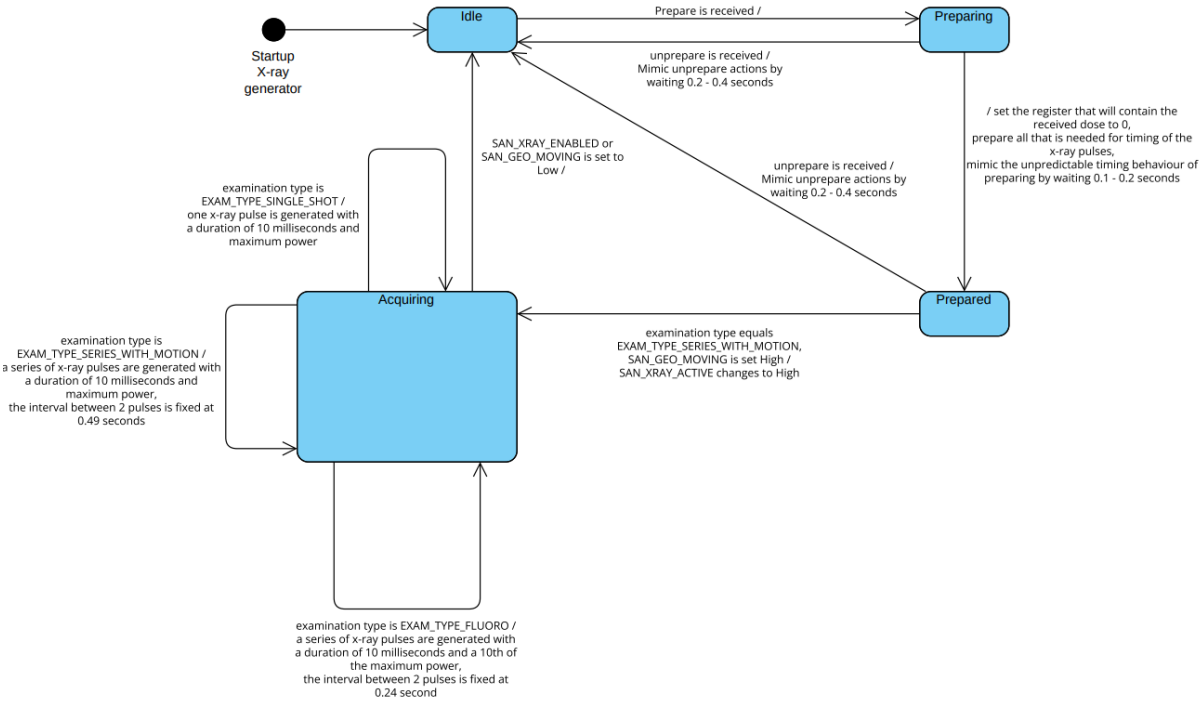
Use-case ID: UC_303	**Use-case:** Data transfer
Description	The actor obtains data from the X-ray generator module
Actor:	CentralAcquisition
Flow:	1. Actor asks to obtain data from the X-ray generator module 2. The X-ray generator module says it is not connected to the actor 3. Else the X-ray generator module says no command was executed that could have generated data 4. Else the X-ray generator module says the data is not a valid data type to be transferred 5. Else the X-ray generator module transfers data to or from the actor

4.6 XrayGenerator state diagram

In the state diagram for the x-ray generator module the different states it can go in are visible. These state are:

- Idle
- Preparing
- Prepared
- Acquiring

The way these states can change into each other and what happens when changing to the different states is shown in the state diagram below



Picture 18: In this picture the state diagram of the X-ray generator module is visible